FIT5230 Malicious AI

Generative Adversarial Network
&
Game Theory

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Generative Adversarial Network
Quick Recap

Generative Adversarial Networks

Generative Adversarial Nets

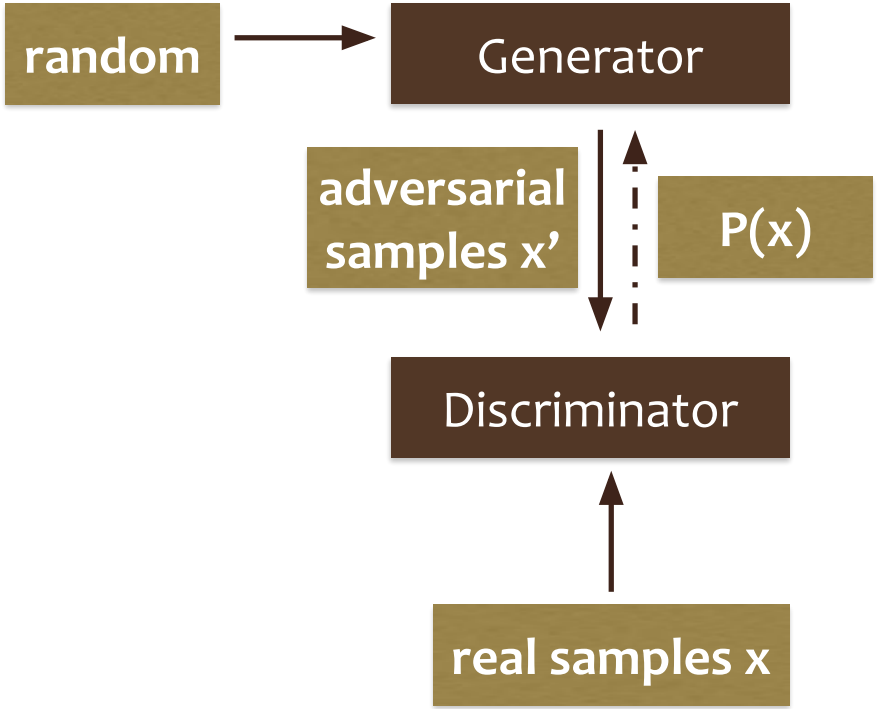
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Abstract

We propose a new framework for estimating generative models via an adversarial process, in which we simultaneously train two models: a generative model G that captures the data distribution, and a discriminative model D that estimates the probability that a sample came from the training data rather than G. The training procedure for G is to maximize the probability of D making a mistake. This framework corresponds to a minimax two-player game. In the space of arbitrary functions G and D, a unique solution exists, with G recovering the training data

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Generative Adversarial Networks



G/D Min-Max Game

The adversarial nature of GANs creates a Min-Max Game

- leading to the final loss function:

$$\min_G \max_D V(D,G) = \mathbb{E}_{\boldsymbol{x}\sim p_{\text{data}}(\boldsymbol{x})}[\log D(\boldsymbol{x})] + \mathbb{E}_{\boldsymbol{z}\sim p_{\boldsymbol{z}}(\boldsymbol{z})}[\log(1 - D(G(\boldsymbol{z})))]$$

- The **discriminator** tries to maximize this function by correctly classifying real and fake data.
- The **generator** tries to minimize it by generating realistic samples that fool the discriminator.

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Game Theory
Prisoner's Dilemma

Overview

- Theory of Games
- Nash Equilibrium
  - Prisoner's Dilemma
- Pure Strategy/Mixed Strategy Nash Equilibrium
  - Matching Coins game
  - Penalty Kick game
  - Battle of the Sexes game

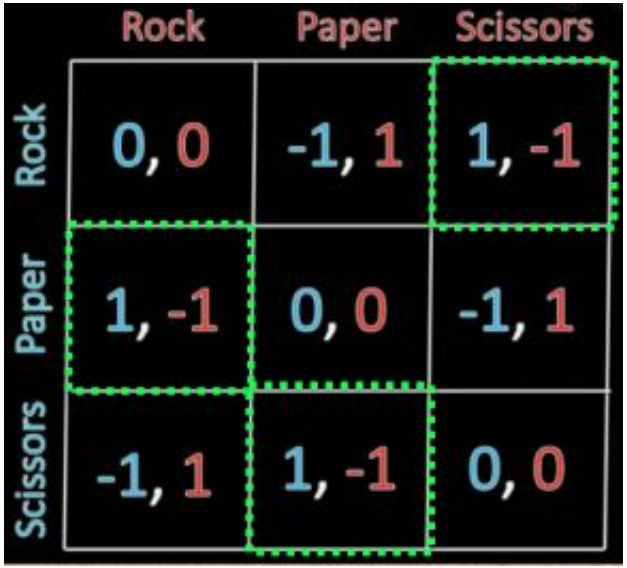
Theory of Game

1. **Players** indexed by $i \in \{1, \ldots, \ell\}$: P_i
2. **Strategies** S_i = set of available actions for player i
   - $s_i \in S_i$ = action by player i
3. **Payoffs** $u_i(S)$ = payoff/utility function of player i
   - where S is list of actions of player i

Note: Payoff depends on all players' strategies

Payoff Matrix

- a_{ij} is payoff in a round
- P1 makes move i, P2 makes move j
  - +ve if P1 wins,
  - 0 if tie,
  - -ve if P1 loses
- zero-sum game
  - sum of winnings of all players = 0

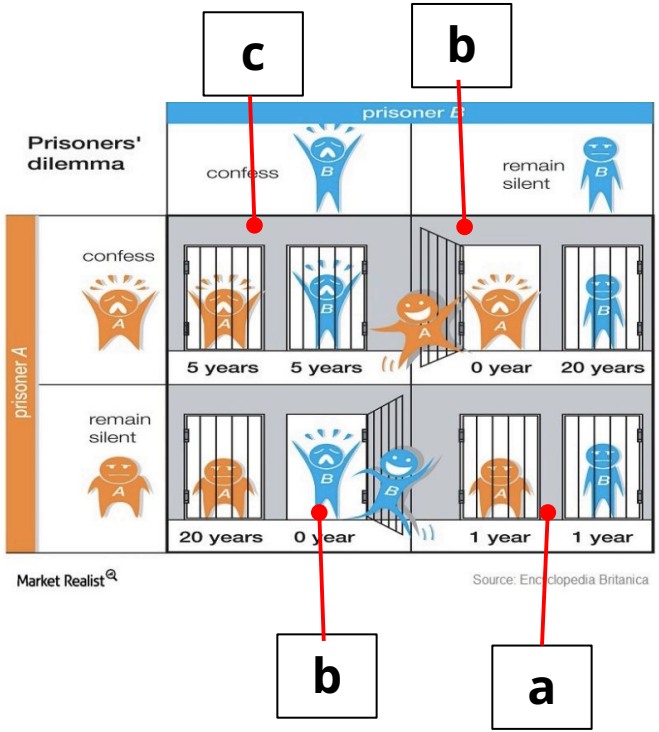


Nash Equilibrium

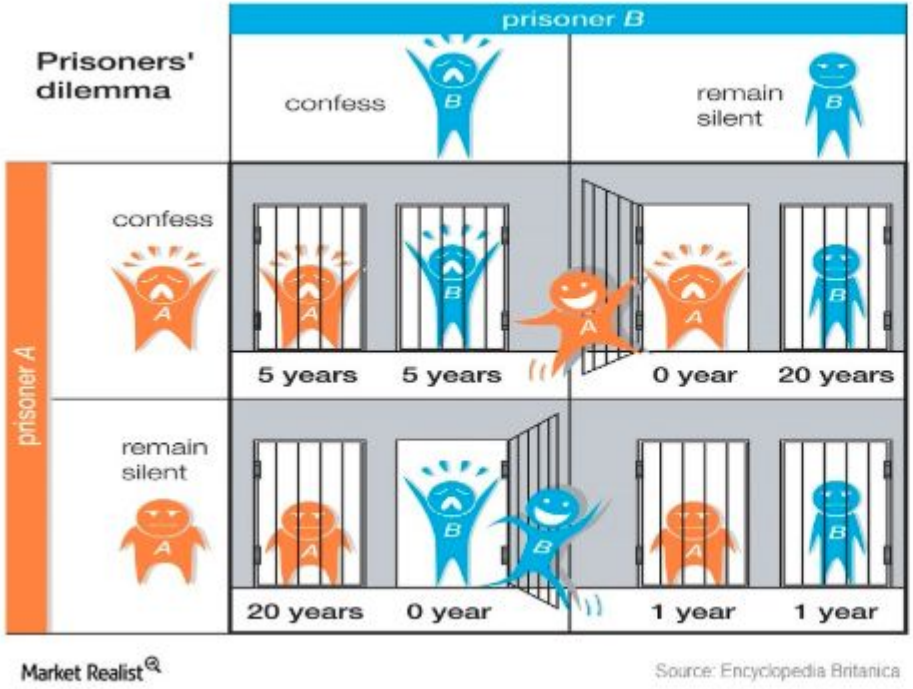
- A concept in game theory where the game reaches an **optimal outcome**.
- This is a state that gives individual players **no incentive to deviate** from their initial strategy.
- The players know their opponent's strategy and still will not deviate from their initial chosen strategies because it remains the **optimal strategy** for each player.
- A game may have **multiple Nash equilibria** or **none** at all.

Eg: Prisoner's Dilemma

1. Two suspects 옷 and 웃 arrested, questioned by police.
   a. If no one confesses: 1 year jail each
   b. if one confesses, the other keeps quiet:
      i. leniency for the rat: 0
      ii. years severe punishment for silence: 20 years
   c. if both confess: both punished equally: 5 years



Prisoner's Dilemma - Payoff Matrix



옷\웃	Confess	Defect
Confess	(–5,–5)	(0,–20)
Defect	(–20,0)	(–1,–1)

Best outcome!

Let's look from 옷 perspective:

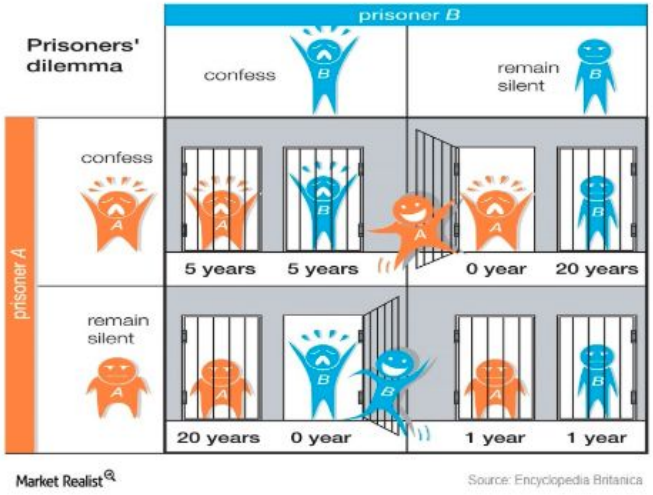
옷\읏	Confess	Defect
Confess	(–5,–5)	(0,–20)
Defect	(–20,0)	(–1,–1)

1. If 옷 confesses:
   a. confess gets 옷 5 years
   b. defects gets 옷 20 years

2. If 옷 defects:
   c. confess gets 옷 0 years
   d. defects gets 옷 1 year

3. Regardless of 옷 confesses/defects:
   e. Confesses gets 옷 5 or 0 years
   f. Defects get 옷 20 or 1 years

Prisoner's Dilemma - Payoff Matrix



옷\웃	Confess	Defect
Confess	(–5,–5)	(0,–20)
Defect	(–20,0)	(–1,–1)

Not stable, not equilibrium → **Best outcome!**

Let's look from 옷 perspective:

옷\웃	Confess	Defect
Confess	(–5,–5)	(0,–20)
Defect	(–20,0)	(–1,–1)

Optimal outcome!
(Stable)

1. If 옷 confesses:
   a. confess gets 옷 5 years
   b. defects gets 옷 20 years

2. If 옷 defects:
   c. confess gets 옷 0 years
   d. defects gets 옷 1 year

3. Regardless of 옷 confesses/defects:
   e. Confesses gets 옷 5 or 0 years
   f. Defects get 옷 20 or 1 years

Types of Nash equilibrium

1. Pure Strategy Nash Equilibrium
- each player only one (pure) strategy: {Confess,Defect}
2. Mixed Strategy Nash Equilibrium
- Nash's Existence Theorem: every game with a finite number of players who can choose from finitely many pure strategies, has at least one Nash equilibrium
- Probability distribution over multiple pure strategies: each one may be chosen based on some probability

Nash Equilibrium

Stable outcome/state,

- where no player can gain any better payoff by changing his/her strategy
- given that other players have chosen their action

Pure strategy Nash Equilibrium

- each player only one (pure) strategy: {Confess,Defect}

옷\옷	Confess	Defect
Confess	(–5,–5)	(0,–20)
Defect	(–20,0)	(–1,–1)



Non-cooperative games

J Nash - Annals of mathematics, 1951 - JSTOR

… prove that a finite **non-cooperative game** always has at least … **non-cooperative game** and prove a theorem on the geometrical structure of the set of equilibrium points of a solvable **game**. …

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Nash Equilibrium

Pure strategy Nash Equilibrium

- each player only one (pure) strategy
- e.g. H or T, L or R, Go or Stop, Confess or Defect

Q: What if can't achieve pure strategy Nash Equilibrium? but one must exist …

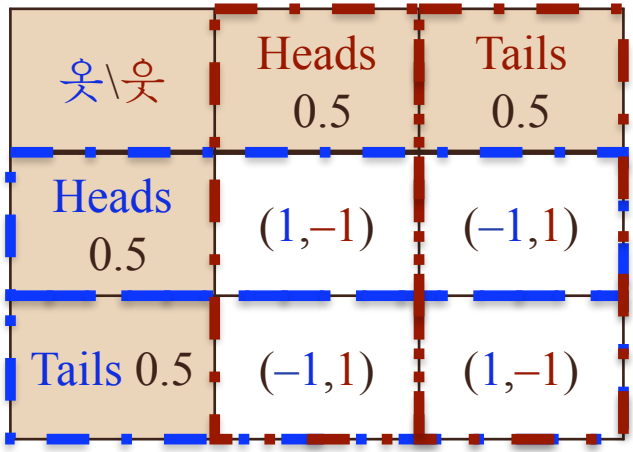
- **Nash's Existence Theorem**
  - every game with a finite number of players who can choose from finitely many pure strategies, has at least one Nash equilibrium

Based on Nash's Existence Theorem, if no equilibrium for pure strategies, must have for mixed

Mixed Strategy Nash Equilibrium

Mixed strategy

- each player chooses an action with probability 0.5
- Payoff for each player for choosing an action:
  - payoff옷(H) = 0.5(1)+0.5(–1) = 0
  - payoff옷(T) = 0.5(–1)+0.5(1) = 0
  - payoff웃(H) = 0.5(–1)+0.5(1) = 0
  - payoff웃(T) = 0.5(1)+0.5(–1) = 0



vs pure strategy

- payoff either 1 or –1

Examples

Strategies: {Heads, Tails} / {Left, Right}

Pure Strategy Nash Equilibrium: which one is the best strategy?

옷\옷	Heads	Tails
Heads	(1,–1)	(–1,1)
Tails	(–1,1)	(1,–1)

Matching Coins Game

옷\옷	Left	Right
Left	(1,–1)	(–1,1)
Right	(–1,1)	(1,–1)

Penalty Kick Game

Battle of the Sexes game

- 2 players: man 웃 and woman 웃
  - get together for night for entertainment, but no communication
  - actions: choose ballet or watch fight
  - man 웃 prefers fight, woman 웃 prefers ballet
  - both prefer together vs alone

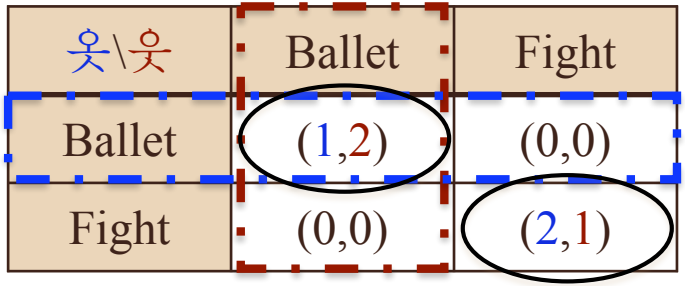
Payoff matrix

웃\웃	Ballet	Fight
Ballet	(1,2)	(0,0)
Fight	(0,0)	(2,1)

Battle of the Sexes game

Pure strategy Nash equilibrium **exists**

- since 웃 chose Ballet
  옷: no point to change from Ballet to Fight
- since 옷 chose Ballet
  옷: no point to change from Ballet to Fight
  …



Pure strategy Nash equilibrium:

- <Ballet, Ballet> or
- <Fight, Fight>

Mixed Strategy Algorithm

Solve for 옷's mixed strategy:

① Target: payoff웃(Ballet) = payoff웃(Fight)
- so it won't matter to 웃 which one she choses

② payoff웃(B)= P옷(B)(2) + (1-P옷(B))(0)

③ payoff웃(F)= P옷(B)(0) + (1-P옷(B))(1)
- where P옷(B) is probably of 옷 choosing Ballet

옷\웃	Ballet	Fight
Ballet	(1,**2**)	(0,0)
Fight	(0,**0**)	(2,1)

Mixed Strategy Algorithm

Solve for 옷's mixed strategy:

① Target: payoff웃(Ballet) = payoff웃(Fight)
- so it won't matter to 웃 which one she choses

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- where P옷(B) is probably of 옷 choosing Ballet

옷\웃	Ballet	Fight
Ballet	(1,2)	(0,0)
Fight	(0,0)	(2,1)

Mixed Strategy Algorithm

Solve for 옷's mixed strategy:

P옷(B)(2) + (1-P옷(B))(0) = P옷(B)(0) + (1-P옷(B))(1) → Target ①

2P옷(B) = (1-P옷(B))(1)
2P옷(B) = 1-P옷(B)
3P옷(B) = 1
P옷(B) = ⅓ → Probability if 옷 choosing Ballet
1-P옷(B) = ⅔ → Probability of 옷 choosing Fight

Mixed Strategy Algorithm

Solve for 웃's mixed strategy:

① Target: payoff옷(Ballet) = payoff옷(Fight)
- so it won't matter to 옷 which one he choses

② **payoff옷(B)= P웃(B)(1) + (1-P웃(B))(0)**

③ payoff옷(F)= P웃(B)(0) + (1-P웃(B))(1)
- where P웃(B) is probably of 웃 choosing Ballet

옷\웃	Ballet	Fight
Ballet	(1,2)	(0,0)
Fight	(0,0)	(2,1)

Mixed Strategy Algorithm

Solve for 웃's mixed strategy:

① Target: payoff옷(Ballet) = payoff옷(Fight)
- so it won't matter to 옷 which one he choses

② payoff옷(B)= P웃(B)(1) + (1-P웃(B))(0)

③ payoff옷(F)= P웃(B)(0) + (1-P웃(B))(2)
- where P웃(B) is probably of 웃 choosing Ballet

옷\웃	Ballet	Fight
Ballet	(1,2)	(0,0)
Fight	(0,0)	(2,1)

Mixed Strategy Algorithm

Solve for 웃's mixed strategy:

P웃(B)(1) + (1-P웃(B))(0) = P웃(B)(0) + (1-P웃(B))(2) → Target ①

P웃(B) = (1-P웃(B))(2)
P웃(B) = 2-2P웃(B)
3P웃(B) = 2
P웃(B) = ⅔ → Probability if 웃 choosing Ballet
1-P웃(B) = ⅓ → Probability of 웃 choosing Fight

Mixed Strategy Algorithm

What we have:

- P옷(B) = ⅓ → Probability if 옷 choosing Ballet
- 1-P옷(B) = ⅔ → Probability of 옷 choosing Fight
- P웃(B) = ⅔ → Probability if 웃 choosing Ballet
- 1-P웃(B) = ⅓ → Probability of 웃 choosing Fight

Multiply the probability of each decision to get the final probability for that outcome.

옷\웃	Ballet (⅔)	Fight (⅓)
Ballet (⅓)	(1,2) P=2/9	(0,0) P=□
Fight (⅔)	(0,0) P=4/9	(2,1) P=2/9

Mixed Strategy Algorithm

Payoff for each player for choosing an action:

① compute the P of each outcome, using P옷 and P웃
② for each outcome, multiply P by player's payoff, & sum these up for all outcomes

payoff옷 = 2/9(1) + 1/9(0) + 4/9(0) + 2/9(2)= ⅔
payoff웃= 2/9(2) + 1/9(0) + 4/9(0) + 2/9(1) = ⅔

Pure strategy payoffs are better; just give in & follow

옷\웃	Ballet (⅔)	Fight (⅓)
Ballet (⅓)	(1,2) P=2/9	(0,0) P=□
Fight (⅔)	(0,0) P=4/9	(2,1) P=2/9

GAN as a Game

A **Generator (G)** and a **Discriminator (D)** are competing in a game.

- The Generator's goal is to create fake samples that look real.
- Discriminator goal is to correctly distinguish real samples from fake ones.

This is a **two-person game**.
The players are Generator (G) and Discriminator (D).

It is also a **zero-sum game**.
This means that the Generator's gain is the Discriminator loss.

Share your answer here: https://shorturl.at/k6fTL

GAN as a Game

The payoffs are as follows:

- If the Generator produces a fake sample and the Discriminator catches it, the Discriminator gains a payoff of **+1**, while the Generator gains **-1**.
- If the Generator produces a fake sample and the Discriminator is fooled, the Generator gains a payoff of **+1**, while the Discriminator gains **-1**.
- To simplify, in all other cases there is **no gain or loss** for either player.

Share your answer here: https://shorturl.at/k6fTL

GAN Games

What you need to do:

1. Draw the payoff matrix for this game.
2. What is the Nash equilibrium of this game?
3. What is the optimal outcome for both players?
4. Discuss: What happens if one side always wins (e.g., Generator dominates or Discriminator dominates)? How does this relate to GAN training stability?

Share your answer here: https://shorturl.at/k6fTL

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Generative Adversarial Network

Motivation

GANs Summary

- First presented by Ian Goodfellow and associates in 2014
- Composed of two neural networks that cooperate through a competitive process
  - **Generator**: To generate human imperceptible adversarial examples
  - **Discriminator**: To determine real vs fake data

The outcome of this adversarial process is extremely advanced models that can recognize and produce synthetic media that is remarkably realistic.

thispersondoesnotexist.com





Motivation

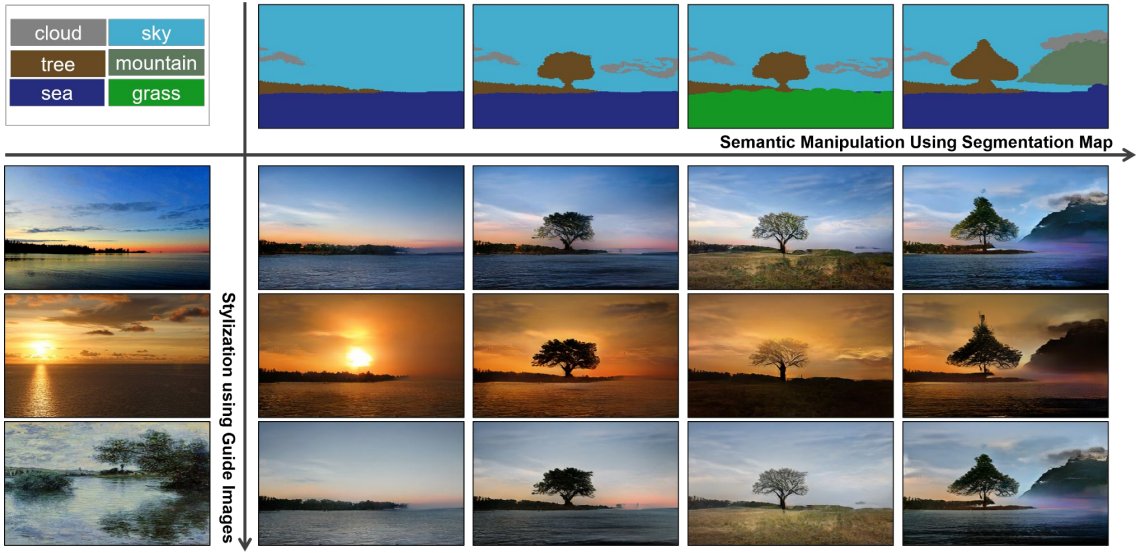
GANs are the state-of-the-art in image generation



Reference: Stroke of Genius: GauGAN Turns Doodles into Stunning, Photorealistic Landscapes, NVIDIA research harnesses generative adversarial networks to create highly realistic scenes., 2019

Motivation

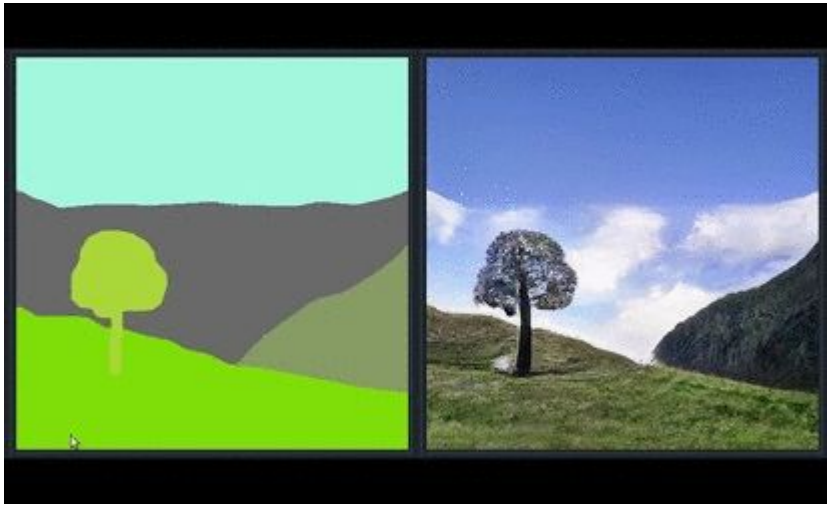
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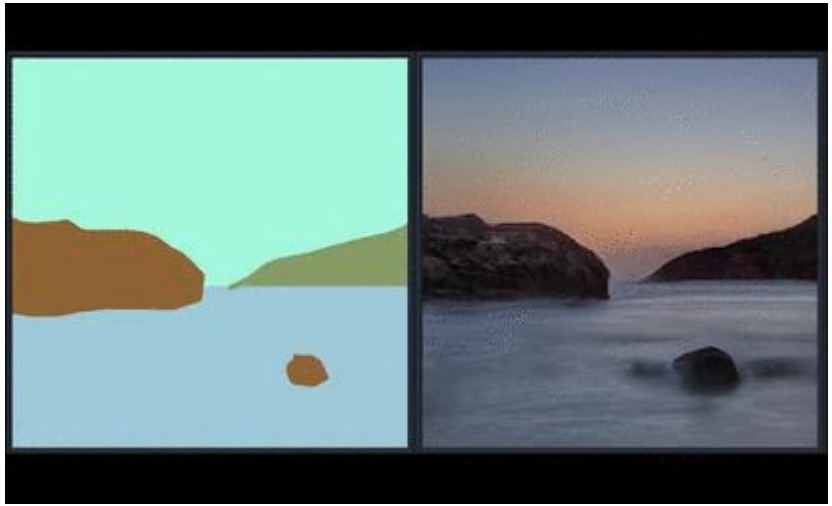


Reference: Semantic Image Synthesis with Spatially-Adaptive Normalization, CVPR 2019
Code Repository: https://github.com/NVlabs/SPADE

Motivation

GANs are the state-of-the-art in image generation



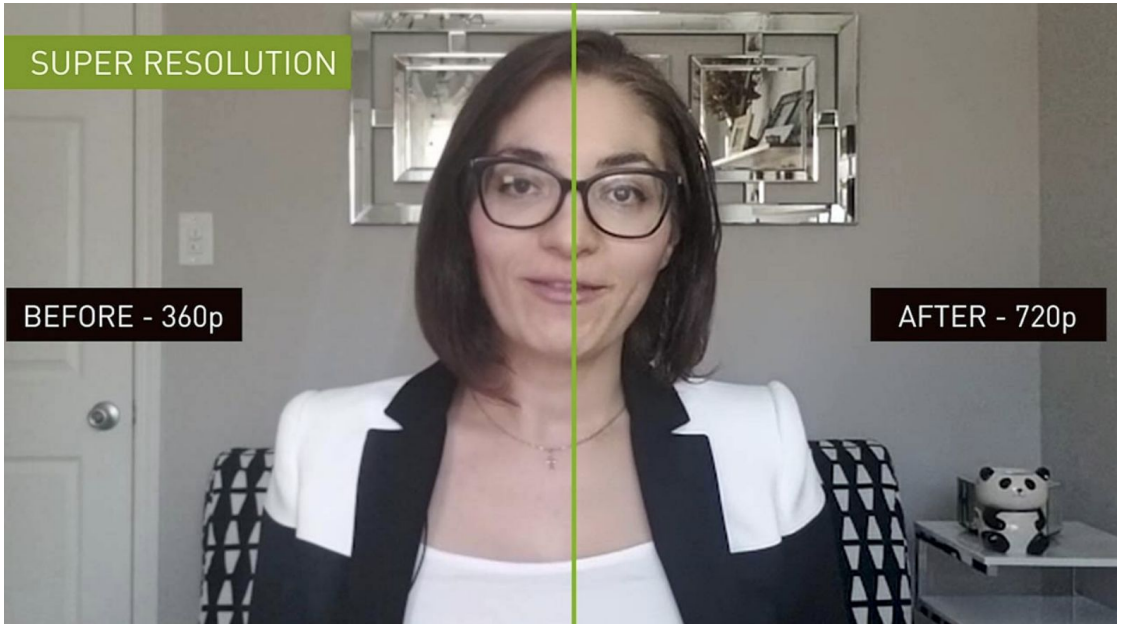


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Motivation

GANs are the state-of-the-art in

- image generation
- video generation

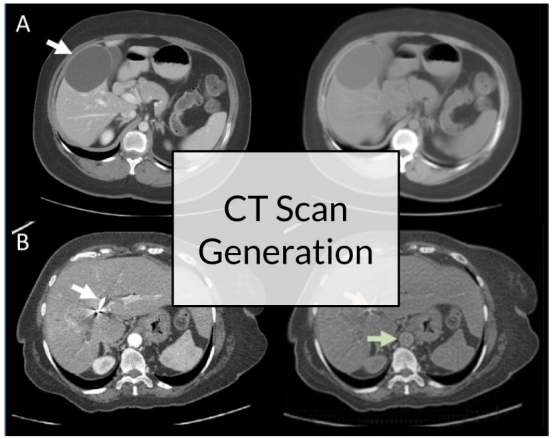


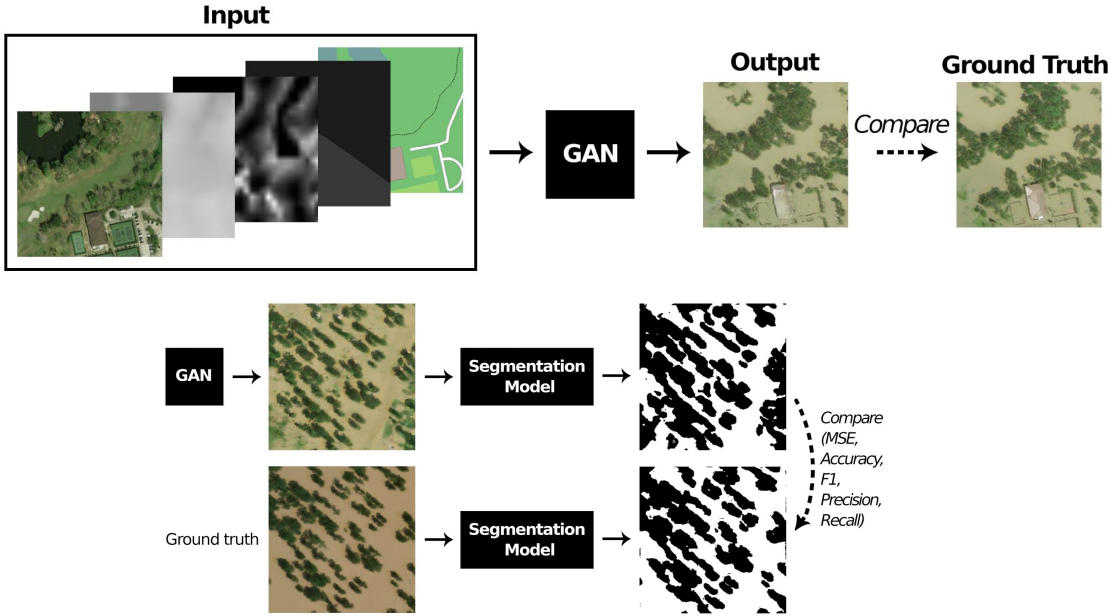
Reference: Photo-Realistic Single Image Super-Resolution Using a Generative Adversarial Network, CVPR 2017
Code Repository: https://github.com/tensorlayer/SRGAN

Motivation

GANs are the state-of-the-art in

- image generation
- video generation





Reference: CCS-GAN: COVID-19 CT Scan Generation and Classification with Very Few Positive Training Images, Generating Aerial Flood Prediction Imagery with Generative Adversarial Networks

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Generative Adversarial Network

AI Security

GANs for Security

The Threat Landscape

- Deepfakes → AI-generated media (video, audio, text, images) that depict fabricated events or speech
- Forged Content → Broader manipulation of digital media beyond deepfakes

Key Risks:

- Misinformation: Undermines trust in news and media.
- Identity Theft & Fraud: Impersonation for financial or personal gain.
- Political Manipulation: Influences elections and public opinion.
- Defamation: Damages reputations through falsified portrayals.

GANs for Security: Threat Detection and Prevention

GANs as a Defense Mechanism

- GANs can be trained on real vs. fake datasets to identify manipulation artifacts.
- Detects subtle inconsistencies in lighting, audio, facial expressions, motion patterns.

Application in Content Moderation

- Automatic Flagging: AI detection for suspicious media.
- Real-Time Analysis: Prevents rapid spread of harmful content.
- User Reporting Integration: Enhances system accuracy with human input.

The Dark Side: GANs as a Security Threat

- Realistic Malware Creation:
  - GANs generate polymorphic malware that evades signature-based antivirus.
  - Subtle perturbations → difficult to classify as malicious.
- Adversarial Evasion Attacks:
  - GANs create adversarial inputs that mislead ML-based security systems.
  - Exploits classifier vulnerabilities.
- Data Poisoning:
  - GAN-synthetic data injected into training pipelines.
  - Corrupts learning process, weakening long-term detection accuracy.

Reference: Bringing a GAN to a Knife-Fight: Adapting Malware Communication to Avoid Detection, Generating Adversarial Malware Examples for Black-Box Attacks Based on GAN, Mal-D2GAN: Double-Detector based GAN for Malware Generation, VagueGAN: A GAN-Based Data Poisoning Attack Against Federated Learning Systems